

Financial Crises

Chapter 12

Econ 351 – Monetary Economics

University of Tennessee, Knoxville

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Preview

- ▶ Financial crises can be explained by **agency problems** and **asymmetric information**.
- ▶ Issues such as **adverse selection** and **moral hazard** can destabilize financial markets and contribute to crises, including the **Global Financial Crisis**.

Road map. Define a financial crisis; trace its three stages (initiation, banking crisis, debt deflation); apply the framework to the Great Depression and the Global Financial Crisis of 2007–2009; and see how regulation responded (macroprudential policy, Dodd–Frank, “too big to fail”).

Outline

What Is a Financial Crisis?

Dynamics of Financial Crises

The Global Financial Crisis of 2007–2009

Response of Financial Regulation

Too-Big-To-Fail and Future Regulation

What Is a Financial Crisis?

A **financial crisis** is a severe disruption in financial markets that prevents funds from being allocated efficiently.

- ▶ Breakdown of information in financial markets
- ▶ Credit markets freeze or become highly constrained
- ▶ Asset prices fall sharply
- ▶ Financial institution and business failures increase

Economic Consequences

Severe financial crises can trigger **recessions** or even **depressions**.

Financial Crisis vs Economic Crisis

Financial Crisis

Problems in the **financial system**:

- ▶ Banks and financial institutions become unstable
- ▶ Credit markets malfunction
- ▶ Funds cannot be allocated efficiently

Economic Crisis

Problems in the **real economy**:

- ▶ Economic output declines
- ▶ Unemployment rises
- ▶ Firms reduce investment and production

Financial crises often **trigger economic crises**, but they are **not the only cause**.

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Dynamics of Financial Crises: Three Stages

Stage 1: Initiation

- ▶ Credit boom and asset-price bubbles
- ▶ Financial liberalization or innovation
- ▶ Increase in uncertainty

Stage 2: Banking Crisis

- ▶ Bank failures and financial instability

Stage 3: Debt Deflation

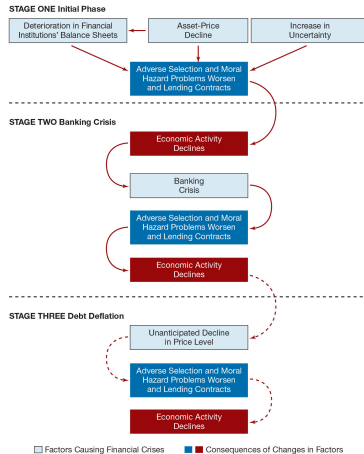
- ▶ Falling prices increase the real burden of debt



Stage 2 in action: depositors queue outside a Northern Rock branch in Birmingham, England, September 2007, in the first run on a British bank in over a century.

Wikimedia Commons, Lee Jordan, CC BY-SA 2.0

Dynamics of Financial Crises



Sequence of events in U.S. financial crises.

Source: Mishkin, Figure 12.1

The Mother of All Financial Crises: The Great Depression

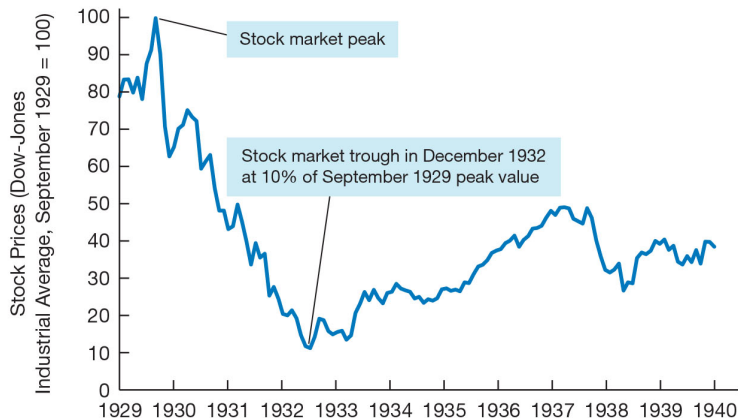
- ▶ The worst economic crisis in U.S. history
- ▶ Lasted from **1929 to the early 1940s**
- ▶ Triggered by a financial collapse of markets, banks, and businesses

Features

- ▶ **Stock Market Crash (1929)** → Investor panic
- ▶ **Bank Panics (1930–1933)** → Massive bank failures
- ▶ **Debt Deflation** → Falling prices increased debt burdens
- ▶ **Global Impact** → Worldwide economic collapse
- ▶ **Rise in Credit Spreads** → Lenders feared default

U.S. unemployment peaked at **25%**, with widespread poverty.

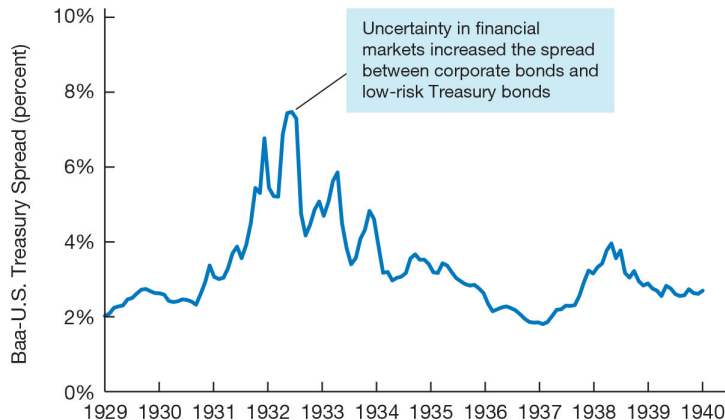
Stock Prices During the Great Depression Period



Stock prices, 1929–1940 (Dow Jones Industrial Average).

Source: Mishkin, Figure 12.2

Credit Spreads During the Great Depression



Spread between Baa corporate bond and Treasury yields, 1919–1941.

Source: Mishkin, Figure 12.3

The Mother of All Financial Crises: The Great Depression

Government Response

- ▶ Initially, the **Federal Reserve did nothing**, which worsened the crisis.
- ▶ **FDR's New Deal (1933–1939):**
 - ▶ **FDIC created** (Federal Deposit Insurance Corporation) to prevent bank runs
 - ▶ **Public works programs** (WPA, CCC) created jobs
 - ▶ **Social Security Act (1935)** established welfare safety nets

Did It Work?

- ▶ **Confidence improved**, but recovery was slow
- ▶ **World War II** ultimately ended the Great Depression by boosting industrial production

The Mother of All Financial Crises: Global Impact

How the Crisis Spread

- ▶ **U.S. demand collapsed** → global trade declined
- ▶ **European banks failed** → economic downturns
- ▶ **Germany and Japan suffered heavily** → rise of extremist regimes

Consequences

- ▶ **International trade collapsed** → global recession deepened
- ▶ **Political instability increased** → contributed to World War II
- ▶ Governments realized the importance of **financial regulation**

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The Global Financial Crisis of 2007–2009

- ▶ The **worst financial crisis** since the Great Depression
- ▶ Originated in the **U.S. housing market** and spread worldwide
- ▶ Led to a **severe global recession** with millions losing jobs and homes

Features

- ▶ **Housing bubble burst** → subprime mortgage defaults increased
- ▶ **Bank failures** → major financial institutions collapsed
- ▶ **Stock market crash** → Dow fell about **50%** (2007–2009)
- ▶ **Global impact** → Europe and other economies experienced deep recessions

Government bailouts and major **financial regulation reforms** were introduced to prevent future crises.

Causes of the Financial Crisis

Main Factors

- ▶ **Mortgage innovation** → rapid growth of subprime MBS/CDOs
- ▶ **Agency problems** → lenders ignored borrower risk
- ▶ **Rating failures** → risky securities rated as safe (AAA)

How It Happened

- ▶ Risky mortgages were **securitized** and widely sold
- ▶ Easy credit fueled the **housing bubble**
- ▶ Ratings understated risk, misleading investors

Overconfidence in financial engineering **masked massive risks.**

Subprime Mortgage Boom & Bust

What Are Subprime Mortgages?

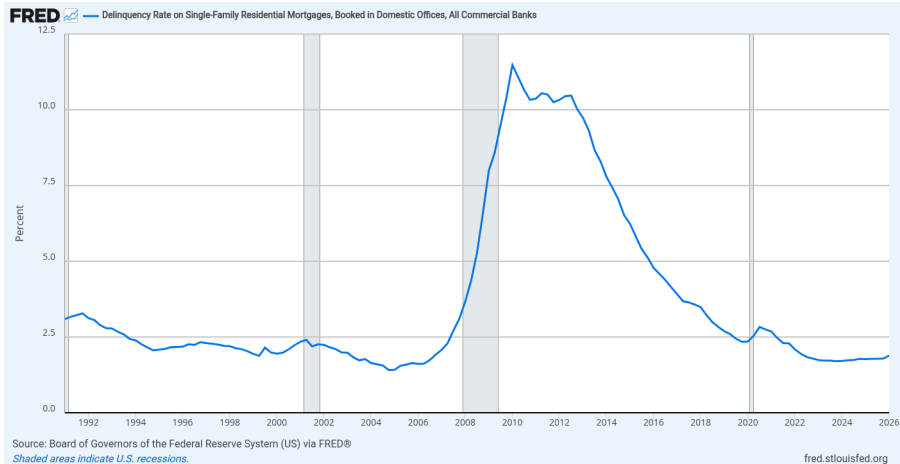
- ▶ Loans to **high-risk** borrowers (poor credit history)
- ▶ Often **adjustable-rate** → payments rose over time
- ▶ Expanded rapidly during the **housing boom (2002–2006)**

What Went Wrong?

- ▶ House prices stopped rising in **2006** → refinancing dried up
- ▶ **Defaults surged** among homeowners
- ▶ Mortgage-backed securities fell sharply → large losses for banks/investors

Mortgage defaults triggered a **chain reaction** in financial markets.

Mortgage Delinquencies Surged After 2006



Delinquency rate on single-family residential mortgages at U.S. commercial banks, quarterly, 1991–2026; the rate rose from under 2% in 2006 to over 11% in 2010.

Source: FRED, Federal Reserve Bank of St. Louis (series DRSFRMACBS)

Financial Engineering and CDOs

What Are CDOs?

- ▶ Banks pooled mortgages into CDOs and sold them to investors
- ▶ **Tranching**: different layers of risk/return
- ▶ **Senior tranche (low risk)** vs. **equity tranche (high risk)**

What Went Wrong?

- ▶ Complexity led investors to **misunderstand risk**
- ▶ Rating agencies gave **AAA** to risky structures
- ▶ When defaults rose, CDO values **collapsed**

CDOs: Quick Recap

CDOs

- ▶ Pool mortgages → create securities (CDOs)
- ▶ Slice into **tranches** (senior vs. equity)
- ▶ Sell widely to investors

Why They Failed

- ▶ Complexity + bad ratings → risk looked “safe”
- ▶ Defaults rose → values collapsed

Financial innovation can **amplify** risk when incentives and information fail.

Credit Rating Agencies' Role

How Did They Contribute?

- ▶ Rated risky mortgage securities as **AAA**
- ▶ **Conflicts of interest**: paid by the firms they rated
- ▶ Understated the true risk of **CDOs and MBS**

Why It Mattered

- ▶ Created **false confidence** in risky assets
- ▶ Encouraged excessive risk-taking by banks and investors
- ▶ Increased **systemic fragility**

When defaults surged, “safe” assets became **worthless**.

Bank Collapses and the Panic (2008)

- ▶ **Mar 2008:** Bear Stearns rescued by JPMorgan
- ▶ **Sep 2008:** Lehman Brothers filed for bankruptcy
- ▶ **Sep 2008:** AIG bailout by U.S. government
- ▶ **Oct 2008:** stock market plunged (about **50%**)

What Went Wrong?

- ▶ **High leverage** made banks fragile
- ▶ **Liquidity crisis:** assets could not be sold quickly
- ▶ **Moral hazard:** risk-taking expecting bailouts



Lehman Brothers headquarters, New York, on the night of 15 September 2008, the day it filed for bankruptcy.

Wikimedia Commons, Robert Scoble, CC

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How the Crisis Spread Worldwide

Transmission Channels

- ▶ U.S. crisis spilled over to **Europe and beyond**
- ▶ European banks held U.S. mortgage securities → heavy losses
- ▶ Governments ran large deficits to support banks

Countries Hit Hardest

- ▶ **Greece (2010)** → sovereign debt crisis and austerity
- ▶ **Ireland, Portugal, Spain** → banking stress and recessions
- ▶ Stock markets and global trade fell sharply

Global financial markets are deeply **interconnected**.

U.S. Response to the Crisis

Federal Reserve Actions

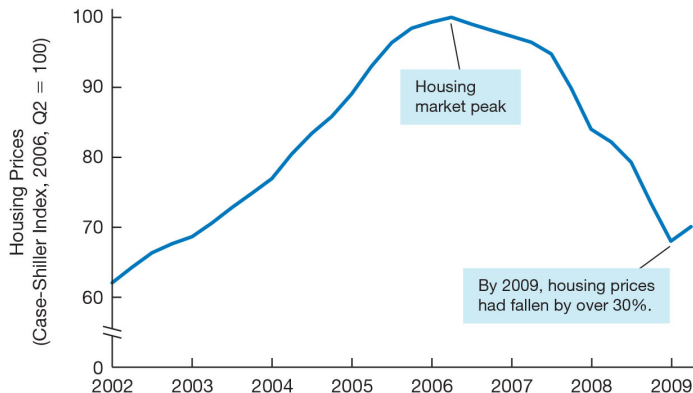
- ▶ Cut policy rates to **near 0%**
- ▶ Provided **emergency liquidity** to financial institutions

Fiscal and Regulatory Actions

- ▶ **TARP (2008)** → capital support for banks (about \$700B)
- ▶ **FDIC insurance expanded** → reduced bank-run risk
- ▶ **Stimulus (2009)** → fiscal support (about \$787B)

Recovery was slow, but policies helped avoid a full-scale depression.

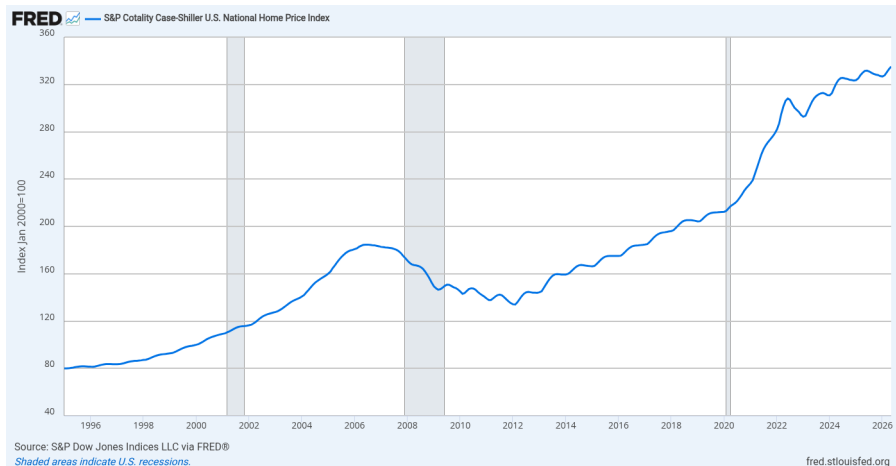
Housing Prices and the Financial Crisis of 2007–2009



Case-Shiller U.S. house price index, 2000–2010.

Source: Mishkin, Figure 12.4

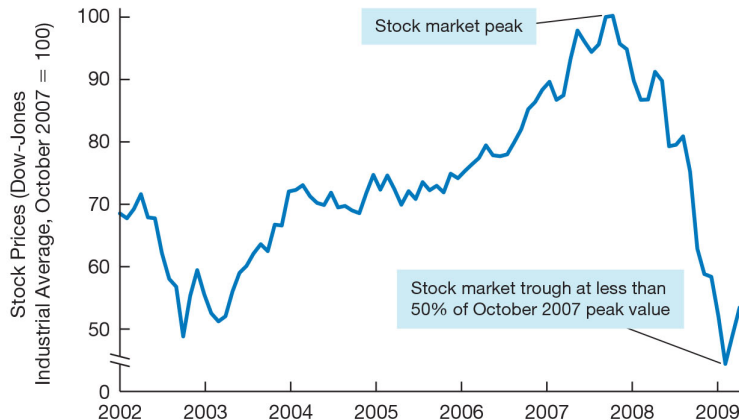
House Prices: The Longer View



S&P CoreLogic Case-Shiller U.S. National Home Price Index (Jan 2000 = 100), 1995–2026: the 2002–2006 boom, the 2006–2012 bust, and the later recovery.

Source: *FRED*, Federal Reserve Bank of St. Louis (series *CSUSHPINSA*)

Stock Prices and the Financial Crisis of 2007–2009



Dow Jones Industrial Average, 2007–2009.

Source: Mishkin, Figure 12.5

Was the Fed Responsible for the Housing Bubble?

Argument: The Fed Contributed to the Bubble (Taylor)

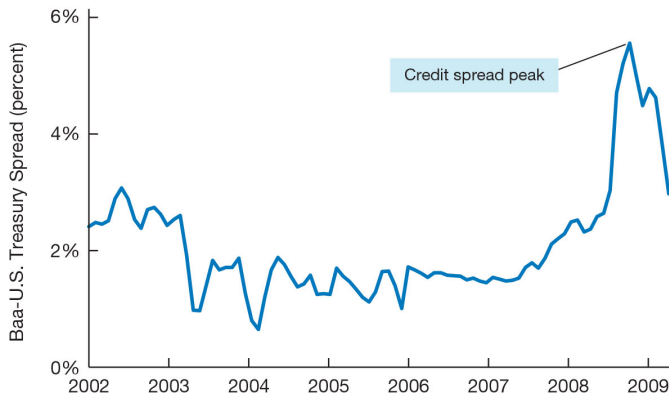
- ▶ The Fed kept interest rates **too low (2003–2006)**
- ▶ Low policy rates → **low mortgage rates**
- ▶ Cheap credit → higher housing demand
- ▶ Expansion of **subprime mortgages**

Counterargument: Other Factors Were More Important (Bernanke)

- ▶ Financial innovation in mortgage products
- ▶ Relaxed lending standards
- ▶ Large capital inflows from emerging economies

Economists still debate whether monetary policy played a major role in the housing bubble.

Credit Spreads and the 2007–2009 Financial Crisis



Spread between Baa corporate bond and Treasury yields, 2007–2009.

Source: Mishkin, Figure 12.6

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Response of Financial Regulation

Before the Crisis: Microprudential Regulation

- ▶ Focused on **individual financial institutions**
- ▶ Ensured banks met **capital and disclosure requirements**
- ▶ Largely ignored **systemic risks** across the financial system

After the Crisis: Macroprudential Approach

- ▶ Focus on **system-wide financial stability**
- ▶ Aim: prevent **financial contagion**

New Policy Tools

- ▶ **Countercyclical capital buffers** (capital raised in booms, relaxed in recessions)
- ▶ **Liquidity regulations** (to ensure stable bank funding)

Dodd–Frank Financial Reform (2010)

The **Dodd–Frank Wall Street Reform Act** was the most significant U.S. financial regulation since the Great Depression.

Reforms

- ▶ **Consumer Financial Protection Bureau (CFPB)**
- ▶ **Bank stress tests** for large banks
- ▶ **Orderly liquidation** of failing financial firms (ending “Too Big To Fail” expectations)
- ▶ **Volcker Rule** restricting proprietary trading by banks
- ▶ Stronger oversight of **derivatives markets**
- ▶ Limits on the Fed’s emergency lending powers

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Too-Big-To-Fail and Future Regulation

Too Big To Fail

- ▶ Very large financial institutions create **systemic risk**
- ▶ Their failure could destabilize the entire economy
- ▶ Government bailouts may create **moral hazard**

Possible Solutions

- ▶ **Break up large banks**
- ▶ **Higher capital requirements**
- ▶ Stronger financial regulation (e.g., Dodd–Frank, Volcker Rule)

Financial regulation aims to reduce systemic risk while maintaining financial stability.

Chapter Summary

- ▶ A financial crisis is a severe disruption of financial markets in which asymmetric-information problems (adverse selection, moral hazard) stop funds from flowing to productive uses.
- ▶ Crises unfold in stages: initiation (credit boom, asset bubble, uncertainty), banking crisis, and debt deflation.
- ▶ The Great Depression: stock crash, bank panics of 1930–33, debt deflation, 25% unemployment; the Fed initially did nothing and the New Deal created the FDIC.
- ▶ The 2007–2009 crisis began with subprime mortgages, securitization (MBS, CDOs), and rating failures; Bear Stearns, Lehman, and AIG marked the panic, and TARP, near-zero rates, and stimulus followed.
- ▶ Regulation shifted from microprudential to macroprudential (countercyclical capital buffers, liquidity rules, Dodd–Frank), though “too big to fail” remains unresolved.